

Section [Section-A] 1 of 2

Question : 1 of 5

Marks for this Question : 1



Select the correct answer

Which of the following represents correct order of subsumption amongst logic coverage criteria? In the options below read $\rightarrow$ as "subsumes".

- ☐ Combinatorial coverage $\rightarrow$ GACC $\rightarrow$ clause coverage.
- ☐ Combinatorial coverage $\rightarrow$ GACC $\rightarrow$ predicate coverage.

Finish 

This is the beginning of the test!

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Section [Section-A] 1 of 2

Question : 3 of 5

Marks for this Question : 1



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Select the correct answer

Save

State true or false: We group du-paths with respect to a variable by its uses.

☐ True

☐ False

Flag

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Section [Section-A] 1 of 2

Question : 4 of 5

Marks for this Question : 1



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Select the correct answer

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Which of the following represents a correct order of subsumption among graph coverage criteria? In the options below, read $\rightarrow$ as "subsumes".

- ☐ Prime path coverage $\rightarrow$ edge pair coverage $\rightarrow$ node coverage.
- ☐ Complete path coverage $\rightarrow$ prime path coverage $\rightarrow$ edge pair coverage.
- ☐ All du paths coverage $\rightarrow$ prime path coverage $\rightarrow$ edge coverage.
- ☐ Prime path coverage $\rightarrow$ all du paths coverage $\rightarrow$ edge coverage.

Finish

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Section [Section-A] 1 of 2

Question : 2 of 5

Marks for this Question : 1



Select the correct answer

In a control flow graph, a path that begins in an initial vertex and ends in a final vertex is called as _____

- ☐ A feasible test path.
- ☐ An infeasible test path.
- ☐ A test path.
- ☐ A test path that tours a final vertex.

Finish

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Start Test

Select the correct answer

Consider the predicate $p = a \vee b \vee (c \wedge d)$. Answer the following question with reference to logic coverage criteria on this predicate. Which of the following options identifies pairs of rows from your truth table that satisfy General Active Clause Coverage (GACC) for clauses a and c ? The numbers in the options below are the row numbers from the truth table that you should have written for the question above.

- ☐ GACC pairs for clause a are {6, 7, 8} x {14, 15, 16} and GACC pair for clause c is {13, 15}
- ☐ GACC pairs for clause a are {1, 2, 3, 4, 5} x {9, 10, 11, 12, 13} and GACC pair for clause c is {13, 15}

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Select the correct answer

For a predicate p and a clause c in p , if p_c evaluates to false, then which of the following holds true?

- ☐ ACC criteria are infeasible for p with respect to c .
- ☐ ICC criteria are infeasible for p with respect to c .

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Select the correct answer

Consider the predicate $p = a \vee b \vee (c \wedge d)$. Answer the following question with reference to logic coverage criteria on this predicate. Which of the following options is true regarding Correlated Active Clause Coverage (CACC) for clauses a and c ?

- ☐ CACC pairs for clauses a and c are different from GACC pairs for these two clauses respectively.
- ☐ CACC pairs for clauses a and c are the same as GACC pairs for these two clauses respectively.

Finish

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This is the end of the test!

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Select the correct answer

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General Instructions

Successfully saved.

Consider the predicate $p = a \vee b \vee (c \wedge d)$. Answer the following question with reference to logic coverage criteria on this predicate. Which of the following options identifies Restricted Active Clause Coverage (RACC) for the clauses a and c ?

- ☐ RACC pairs for the clause c are the same as GACC pairs for this clause. RACC pairs for clause a are (6, 14), (7, 15) and (8, 16)
- ☐ RACC pairs for the clause c are the same as CACC pairs for this clause. RACC pairs for clause a is (6, 14) only
- ☐ RACC pairs for the clause c are the same as CACC pairs for this clause. RACC pairs for clause a are (6, 14), (7, 15) and (8, 16)
- ☐ RACC pairs for the clause c are the same as GACC pairs for this clause. RACC pairs for clause a are (6, 14) and (7, 15)

Finish

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Next (2)

Section [Section-B] 2 of 2

Question : 6 of 10

Marks for this Question : 2



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General Instructions

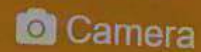
Successfully saved.

Use the following methods `trash()` and `takeOut()` given below to answer the following question:

```
1 public void trash (int x)    15 public int takeOut (int a, int b)
2 {                            16 {
3     int m, n;                17     int d, e;
4                               18
5     m = 0;                    19     d = 42*a;
6     if (x > 0)                 20     if (a > 0)
7         m = 4;                21         e = 2*b+d;
8     if (x > 5)                 22     else
9         n = 3*m;              23         e = b+d;
10    else                       24     return (e);
11        n = 4*m;              25 }
12    int o = takeOut (m, n);
13    System.out.println ("%o is: " + o);
14 }
```

Provide test inputs (in terms of values for x) for all-coupling-uses criterion. If there is any criterion that is not satisfiable or feasible for a coupling variable pair, state that instead of providing a test input.

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General Instructions

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Consider a graph given by $G = (N, E)$ where the set of vertices $N = \{1, 2, 3, 4, 5, 6\}$, with initial vertex $N_0 = \{1\}$, final vertex $N_f = \{6\}$ and the set of edges $E = \{(1, 2), (2, 3), (2, 6), (3, 4), (3, 5), (4, 5), (5, 2)\}$. The question deals with data flow coverage criteria for a variable x on this graph with $def(x) = \{1, 3\}$, $use(x) = \{3, 6\}$. Assume that the use of x in 3 precedes the def .

Consider the test path $t = [1, 2, 3, 4, 5, 2, 3, 5, 2, 6]$. List all the du -paths toured by this test path, please include direct touring and tours with side-trips, as applicable

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Section [Section-B] 2 of 2

Question : 4 of 10

Marks for this Question : 2



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General Instructions



Saved successfully.

Consider a graph given by $G = (N, E)$ where the set of vertices $N = \{1, 2, 3, 4, 5, 6\}$, with initial vertex $N_0 = \{1\}$, final vertex $N_f = \{6\}$ and the set of edges $E = \{(1, 2), (2, 3), (2, 6), (3, 4), (3, 5), (4, 5), (5, 2)\}$. The question deals with data flow coverage criteria for a variable x on this graph with $def(x) = \{1, 3\}$, $use(x) = \{3, 6\}$. Assume that the use of x in 3 precedes the def.

List all the du-paths with respect to x .

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Next (1)

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Consider the predicate $p = a \vee b \vee (c \wedge d)$. Answer the following question with reference to logic coverage criteria on this predicate. Compute and simplify the conditions under which clauses a and c determine the predicate p , i.e., p_a and p_c , respectively.

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Section [Section-B] 2 of 2

Question : 5 of 10

Marks for this Question : 2



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General Instructions



Successfully saved.

Consider a graph given by $G = (N, E)$ where the set of vertices $N = \{1, 2, 3, 4, 5, 6\}$, with initial vertex $N_0 = \{1\}$, final vertex $N_f = \{6\}$ and the set of edges $E = \{(1, 2), (2, 3), (2, 6), (3, 4), (3, 5), (4, 5), (5, 2)\}$. The question deals with data flow coverage criteria for a variable x on this graph with $\text{def}(x) = \{1, 3\}$, $\text{use}(x) = \{3, 6\}$. Assume that the use of x in 3 precedes the def.

List a minimal set of test paths that satisfies all-uses coverage criterion with respect to x .

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General Instructions

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Use the following methods **trash()** and **takeOut()** given below to answer the following question.

```
1 public void trash (int x)      15 public int takeOut (int a, int b)
2 {                               16 {
3     int m, n;                  17     int d, e;
4                                 18
5     m = 0;                      19     d = 42*a;
6     if (x > 0)                  20     if (a > 0)
7         m = 4;                  21         e = 2*b+d;
8     if (x > 5)                  22     else
9         n = 3*m;                23         e = b+d;
10    else                        24     return (e);
11        n = 4*m;                25 }
12    int o = takeOut (m, n);
13    System.out.println ("o is: " + o);
14 }
```

For each pair of coupling variables, list all the last-defs and first-uses. Your answer should list the method names, the coupling variables and the line numbers of the last-def and the first-use respectively.

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Use the following methods `trash()` and `takeOut()` given below to answer the following question.

```
1 public void trash (int x)
2 {
3     int m, n;
4
5     m = 0;
6     if (x > 0)
7         m = 4;
8     if (x > 5)
9         n = 3*m;
10    else
11        n = 4*m;
12    int o = takeOut (m, n);
13    System.out.println ("o is: " + o);
14 }
```

```
15 public int takeOut (int a, int b)
16 {
17     int d, e;
18
19     d = 42*a;
20     if (a > 0)
21         e = 2*b+d;
22     else
23         e = b+d;
24     return (e);
25 }
```

For each pair of coupling variables, list all the last-defs and first-uses. Your answer should list the method names, the coupling variables and the line numbers of the last-def and the first-use respectively.

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